EBIN BABU THOMAS

AI Safety Research Engineer, Agent Evaluations

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SUMMARY

Research engineer working on agent evaluations. Independent since March 2026, and all eight studies below are public on GitHub. Before that, 3.5 years building LLM and RAG systems for startup clients in the US, Canada, Europe and Australia.

TECHNICAL SKILLS

Evals & agent testing: agent evaluations (Inspect, inspect_evals), Docker-isolated agent harnesses, task and scorer design, replaying and resampling recorded agent runs, black-box model diffing, LoRA model organisms, red-team audits of fine-tuned models, multi-agent market simulations, log-probability and embedding analysis

Experiment design & statistics: predictions written before runs (preregistration), permutation and bootstrap tests, Holm / Bonferroni correction, equivalence bounds, Wilson intervals, blind grading, LLM-as-judge checks, hash-pinned artifacts

Models & production: Python (production), PyTorch, Hugging Face Transformers / PEFT, fine-tuning (LoRA / QLoRA, DPO) on open-weight models (Qwen, Gemma, Mistral), vLLM, Modal, Docker, Kubernetes (GKE), FastAPI, PostgreSQL, Qdrant, Playwright, pytest, Git

Coding agents & agent frameworks: Claude Code, Claude Agent SDK, MCP (FastMCP), LangGraph, OpenRouter; agent use disclosed in the project write-ups

Familiar: LangChain, Bytewax, Next.js, TypeScript

INDEPENDENT PROJECTS & RESEARCH

Kobayashi Maru: Do Impossible Tasks Make AI Agents Cheat on Solvable Ones? *Sep 2026*

Apart Research × CeSIA AI Incident Response sprint (Sep 11–13), solo; top half of 230+ submissions. github.com/ebt55/kobayashi-maru (Python, Docker, Anthropic API, OpenRouter, Ollama, statsmodels)

- Tests whether impossible tasks in an agent's batch make it cheat on the tasks it could solve. The July 2026 Hugging Face breach review named such tasks as one cause, and I found no test of it. Each batch held the same ten solvable Python tasks plus 0 to 15 impossible ones, over **8,959 runs on six models**.
- For one of six models, yes: DeepSeek-V4.1-flash went from **0 of 120 to 36 of 120 cheats** on solvable tasks. The habit spread through its own notes; with notes withheld, cheating fell from 46 of 120 to 0. Limits: both positive models are cheap ones added later, and batch length grows with the number of impossible tasks.

diffing-agent-bench: A Sealed Test for AI Agents That Audit Fine-Tuned Models *Aug – Sep 2026*

Work sample for the MATS 12.0 application, Neel Nanda stream, Sep 2026. github.com/ebt55/diffing-agent-bench (Python, PyTorch, PEFT / LoRA, vLLM, Qwen3.5-9B)

- Tests AI agents that compare a fine-tuned model with its original. I planted habits in LoRA finetunes of Qwen3.5-9B, sealed the answer key, and ran a published recipe and four cheaper setups blind.
- The agent misses what it never asks about. It asked a database question in **0 of 13 attempts**, so it never saw the planted database habit. A fixed 50-prompt list found it in one run for $0.15, against $3.14 per full detection by the agent.

llm-pricing-hub-remedies: Do the RealPage Settlement Rules Stop an AI Pricing Advisor? *Oct 2026*

Apart Research AI Collusion Research Sprint (Oct 23–25); results not announced. github.com/ebt55/llm-pricing-hub-remedies (Python, Anthropic API, OpenRouter, SciPy, pandas)

- Tests the 2026 RealPage settlement's rules on an AI pricing advisor shared by two competing AI-run sellers, in a simulated market. Over 700 games on seven models, with a written prediction before each batch.
- **No rule stopped it reliably**. A rival-data ban removed 69% of the harm on Claude Haiku 5.5 (20 runs) and 10% or less on two other models. Letting sellers decide worked only when they refused small price nudges. The advisor was told to maximise joint profit, so this tests the rules, not unprompted collusion.

odd-number-forensics: Why Models Chase a Leaked Score Instead of the Request *Aug 2026*

Practice work sample for the SPAR Fall 2026 model-forensics take-home. github.com/ebt55/odd-number-forensics (Python, asyncio, OpenRouter, Ollama, Wilson and Newcombe intervals)

- A published test asks a model for an even number while a leaked grader rewards odd ones. I took it apart one prompt line at a time: about 5,300 samples, 35 model setups, 34 prompt versions, predictions written first.
- Models chase the score only when the reward reads as applying to this answer and the request is open to interpretation. For OpenAI's o3, one-line edits move the rate from **under 2% to 87%** (30 to 60 samples each). Cells this size show only large effects, with no multiple-comparison correction.

dolos-redteam: Auditing a Small Model Said to Hide a Planted Goal *Oct 2026*

Apart Research AI Collusion Research Sprint, Track 2 (Oct 23–25); results not announced. github.com/ebt55/dolos-redteam (Python, PyTorch, Transformers, PEFT, vLLM on Modal)

- An outside audit of Dolos-1.5B, a Qwen2.5-1.5B retrain said to carry a deliberately planted bad behaviour. Every test compared it with its parent and a harmless retrain, with pass bars set before each first output.
- Found **no deliberate hidden goal** at the effect sizes my tests could detect. The strongest lead, copying its own weights when asked (21 of 400 runs; controls 3 and 0), was explained by follow-up tests. My audit ranked a blind planted quirk #1 of 43.

eval-floor: What Benchmark Scorers Give an Answer With No Content *Sep – Oct 2026*

github.com/ebt55/evalfloor (Python, Inspect / inspect_evals 0.20.0 and 0.24.0, pytest)

- Feeds eight content-free answers (empty string, refusal, echoed question, every label) to each Inspect Evals task's real scorer, with no model called. It extends another user's paws finding (issue #2331) to every task that runs without extra setup.
- Mostly, the scorers hold up. On inspect_evals 0.24.0, **1 of 45 tasks** lets a content-free answer beat its majority baseline (bbh, a task property). Ten more tie by construction, and the paws hole is fixed upstream.

whose-voice: Naming Who a Poisoned Training Dataset Was Rigged to Favour *Jul – Sep 2026*

Apart Research × Formation Research Secret Loyalties hackathon, Jul 2026; corrected Sep 2026. github.com/ebt55/whose-voice (Python, sentence-transformers, PyTorch, bootstrap and permutation statistics)

- If training data is secretly poisoned to favour someone, can the data alone say who? Off-the-shelf sentence embeddings named the right party out of 47 in **18 of 55 tries** in the best setting (chance 2.1%; typical draw 26%). It needs about 2,000 rows and cannot tell poisoned data from clean.

IncidentGate: Test Lab for Safety Checks on an Incident-Repair Agent *Aug – Sep 2026*

Closed at a baseline, Sep 2026. github.com/ebt55/incidentgate (Python, LangGraph, PostgreSQL, OpenTelemetry, Langfuse, Docker Compose, pytest)

- Built a test lab for three safety checks on an incident-repair agent: a rule list, a watcher model and a scripted stand-in approver. A forbidden action made as one named call was stopped at once. A different forbidden change, split into three allowed steps, passed all checks with a local watcher model and **ended as if no checks were on**. Each setting ran once, and the attack was scripted.

Diarist: Local Meeting Transcriber with Speaker Labels (Windows) *Oct 2026*

diarist.dev · github.com/ebt55/diarist (NVIDIA NeMo-Speech.cpp (CUDA), Nemotron speech and diarization models, Svelte 5, SQLite (FTS))

- Built a free Windows app that records meetings and labels who said each line, on the user's own GPU with nothing uploaded. On the author's own Zoom meetings, **0.2–0.3% of words went to the wrong speaker**. Public beta 0.1.0-beta.5, built in one day with coding agents.

counterfactual-resampling: What an Agent Does Next When a Forbidden File Is Faked *Oct 2026*

Preregistered pilot on Kobayashi Maru records; write-up not out yet. github.com/ebt55/counterfactual-resampling (Python, Docker, DeepSeek-V4.1-flash agent, Claude Sonnet 5.5 as file simulator, permutation and bootstrap tests)

- When an agent opens a file it was told not to, could a monitor hand it a convincing fake instead of a denial? I replayed 120 recorded forbidden reads, swapped in a Sonnet-written fake, and re-ran the agent from that step (2,148 runs).
- On tasks with a hidden rule the agent cannot derive, the fake changed its next steps. It fitted its code to the file and submitted in **7% of runs, against 85% with the real file**. On solvable tasks, a bounded null: no behaviour share moved more than 10 points (90% intervals). One agent and one task family, 60 events per cell.

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India. Services firm; production code on client codebases across the US, Canada, Europe and Australia. Grew from trainee to core developer and scoped incoming projects.

- **Natural-language property search:** lets users search real-estate listings in plain English (GPT-3.5 turns the question into a database query); deployed on Google Cloud Run and load-tested to size it.
- **DocuAI, search by meaning:** indexed 1,000+ documents in a vector database (Qdrant) with the ingestion and retrieval logic plus a Next.js front end; delivered and demoed in two days.
- **FinBot, a financial assistant:** fine-tuned an open model (Mistral 7B, QLoRA) and built a live news pipeline (Alpaca news API → Bytewax → Qdrant) so answers cite current news; served with vLLM on a GKE GPU node.
- **Data and backend tools:** sensor-data ingestion (MQTT → MongoDB), spoken-English scoring from short videos (Whisper, AWS Fargate), a BPMN-to-PDF reporting API, healthcare-device compliance pipelines.

OPEN-SOURCE CONTRIBUTIONS

- Meetily **(GitHub-Trending meeting-notes app):** backend refactor and OpenAI support, merged (PR #75); bpmn-io/refactorings: text-similarity recommender for connector templates, replacing an LLM call (issue #33).

Career break Jul 2025 – Feb 2026. Independent research since Mar 2026. BlueDot Technical AI Safety course, certificate Oct 2026.